

Converting images to PDF

Sometimes, you have to create PDF pages from images. ImageMagick is the preferred Linux utility to convert images.

```
# Convert images to PDFs
magick front-cover.png -resize 100% front.pdf
magick back-cover.png -resize 100% back.pdf
```

Concatenating several PDFs

PDFtk is a powerful PDF-processing utility that can perform tasks such as merging, splitting, encrypting, decrypting, stamping, and watermarking.

```
# Concatenate several PDFs
pdftk front.pdf inner-pages.pdf back.pdf \
    output book.pdf
```

Encrypting PDFs

PDF documents can be encrypted using two passwords — *owner* and *user*.

```
pdftk book.pdf output book-encrypted.pdf \
    encrypt_128bit \
    owner_pw RcHrDsTlMn^012 \
    user_pw FrSfTwRfNdtn^321
```

If you set an empty user password, you can let users view the PDF without a password prompt. You can specify additional restrictions using the *'allow'* option. Possible values are *Printing*, *DegradedPrinting*, *ModifyContents*, *Assembly*, *CopyContents*, *ScreenReaders*, *ModifyAnnotations*, *FillIn* and *AllFeatures*. If you do not specify the *allow* option, none of these features will be available. Well, the PDF standard specifies that these features should not be available. In reality, the restrictions are not strictly implemented by many PDF viewer applications.

Removing PDF password

Use the *input_pw* option to specify the password.

```
pdftk book-encrypted.pdf \
    input_pw RcHrDsTlMn^012 \
    output book-decrypt.pdf
```

If you start this command with a space, then the password (RcHrDsTlMn^012) will not be stored in the *'bash'* shell history. Another trick is to use an interactive dialog.

```
sPassword=$(zenity --password \
    --title "Decrypt PDF" \
    --text "Type the password")
pdftk book-encrypted.pdf \
    input_pw $sPassword \
```

```
output book-decrypt.pdf
```

If you want a console-only prompt, then disable the input echo.

```
stty -echo
read -p "Type the password: " sPassword
stty echo
```

Converting PDF pages to images

You can use the *pdftoppm* utility to convert PDF pages to images in JPEG and PNG format. You can also specify the pixel density and page range.

```
# Export pages 2 to 12 with 96 dpi
pdftoppm -png -r 96 -f 2 -l 12 book.pdf page
# creates numbered images with prefix 'page'
```

Rasterising a PDF

Sometimes, you cannot give a PDF as is to someone else or put it online. To foil content scrapers (and since *'AI'* is now an unabashed content scraper), it is best to convert even the text to JPEG images. Why JPEG? Because it is lossy.

```
pdftoppm -jpeg -r 96 book.pdf page
magick page*.jpg book-rasterized.pdf
```

Some content scrapers do have the ability to read text from images, but that task requires extra resources. If you set the DPI to 72, it gets more demanding. For further annoyance, stamp it or watermark it.

```
magick watermark.png watermark.pdf
pdftk book-rasterized.pdf \
    stamp watermark.pdf \
    output pages-watermarked.pdf
```

Converting PDF to DjVu

Many PDF books on *Archive.org* are scanned from library copies of physical books. These PDFs are extremely heavy as the scans are usually dense or big images. Ordinary PDF readers on tablets struggle to read such PDFs. My solution is to convert the PDFs to DjVu. (DjVu viewer applications are optimised for reading from images. They load DjVu pages in a snap.)

```
pdf2djvu --dpi=220 \
    --output=tablet.djvu \
    library.pdf
```

Depending on the page size in the PDF, you will have to adjust DPI with the *--dpi* option.

Continued to page....19